MAE3405 Aerospace Propulsion

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Topic 1-2 Problem Set Questions

Question 1

Taking the data for the surface level atmospheric composition of Mars (table A.9 in the first chapter of the notes), calculate:

1. (i) the average atmospheric molecular weight of the atmosphere, noting that table A.9 lists concentrations as volume fractions.
2. (ii) the gas constant (R kJ/kg.K) and
3. (iii) the ratio of specific heats (γ).

Note: For Argon, Cp=0.523 kJ/kg.K and mole weight 39.95 kg/kmol.

[Answer: (i) 43.41 kg/kmole; (ii) 0.1915 kJ/kg.K; (iii) 1.29]

Question 2

An aircraft engine with constant combined mass flow rate of 0.2 kg/s operates in an environment such that 15 kW of heat is lost from its surfaces through radiant and convective processes. Fuel flow is equal to 1/15 of the inlet mass flow, and provides 45 MJ of energy per kg of fuel to the engine. The inlet gas to the engine can be approximated by air at SSL conditions, while the exhaust mass flow into the atmosphere is a mixture of 65% N_2, 15% CO_2 and 20% water (gas) by mass at 900°C. For an inlet pipe diameter of 0.125m and exhaust pipe diameter of 0.15m, determine:

(i) the power output of the engine and

(ii) the engine's thermal efficiency.

Use a reference temperature of T = 25°C for your calculations.

[Answer: (i) 376.7 kW, (ii) 62.8%.]

Question 3

A small gas turbine engine has the following flight conditions. You may assume the exhaust pressure to be atmospheric.

Altitude	6500m
Flight speed Mach number	0.398
Intake diameter	0.300m
Exhaust diameter	0.285m
Exhaust speed	950kph
Exhaust gas temperature	250^0C
Exhaust gas constant	250J/kg.K
Fuel density	850kg/m^3

(a) Determine the ratio of fuel mass flow to air mass flow for the engine. [Ans: 0.0279]
(b) Calculate the net thrust developed by the engine and the thrust specific fuel consumption (in kg/N.h) [Ans: 806.2 N; 0.688 kg/N.h]
(c) What percentage of the gross thrust is used to overcome ram drag? [Ans: 46.1%]

Question 3D

A 5-horsepower electric motor for a light aircraft operates on an alternating current supply generated from the aircraft's batteries using a 3-phase inverter.

The voltage provided to the motor is 440 V AC RMS. When tested under no-load conditions, the motor's speed is 1195 RPM and it draws 1.2 Amps RMS at a power factor of 30% lagging.

When connected to the propeller and at full throttle, the motor draws 6.8 Amps at a power factor of 78% lagging at 1150 RPM.

Find the motor's input power, power loss and efficiency at no-load and full-load conditions. Calculate the speed regulation of the motor.

HINT: The input power for an AC motor, when measured using RMS voltage and current, is : $P_{in} = \sqrt{3}\, V_{RMS}\, I_{RMS}$.

[Ans: 4042 W, 312 W/274.4 W, 92.28%, 3.91%]